MARK SCHEME · TEACHER COPY

10BR · Chapter 7: Algorithm Design and Problem-solving

Cambridge IGCSE Computer Science 0478

Paper 2 Algorithms, Programming and Logic

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

Mark scheme extracts © Cambridge University Press & Assessment.

Total: 75 marks

Question 1 · Feb/March 2023 Paper 22, original Q8 (a),(b) [7 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>
<table>
<tr><th>NumberSales</th><th>Total</th><th>SaleValue</th><th>Average</th><th>OUTPUT</th></tr>
<tr><td>0</td><td>0</td><td></td><td></td><td></td></tr>
<tr><td>1</td><td>5.50</td><td>5.50</td><td></td><td></td></tr>
<tr><td>2</td><td>8.90</td><td>3.40</td><td></td><td></td></tr>
<tr><td>3</td><td>15.15</td><td>6.25</td><td></td><td></td></tr>
<tr><td>4</td><td>19.00</td><td>3.85</td><td></td><td></td></tr>
<tr><td>5</td><td>8.00</td><td>-11.00</td><td></td><td></td></tr>
<tr><td></td><td></td><td>0</td><td>1.6</td><td>Average sale value 1.6</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td></tr>
</table>
One mark for each column NumberSales, Total, SaleValue
One mark for columns Average and OUTPUT
</td><td>4</td></tr>
<tr><td>1(b)</td><td>Error – one mark for identification of error
for example including negative numbers / not differentiation between negative and positive values
Correction
One mark for correct placement and one mark for appropriate action
For example – after the input box insert a decision box to reject negative numbers</td><td>3</td></tr>
</table>

Total: 75 marks

Question 2 · May/June 2023 Paper 23, original Q7 (a),(b) [8 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)</td><td>
One mark per mark point, max six

correct <code>Total</code> column
correct <code>Value</code> column
correct <code>Five1</code> column
correct <code>Five2</code> column
correct <code>Ten1</code> and <code>Ten2</code> columns
correct OUTPUT column

<table>
<tr><th>Total</th><th>Value</th><th>Five1</th><th>Five2</th><th>Ten1</th><th>Ten2</th><th>OUTPUT</th></tr>
<tr><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>5</td><td>1</td><td>1</td><td>0</td><td>0.5</td><td>Rejected</td></tr>
<tr><td></td><td>50</td><td>10</td><td>10</td><td>5</td><td>5</td><td></td></tr>
<tr><td>50</td><td>52</td><td>10</td><td>10.4</td><td></td><td></td><td>Rejected</td></tr>
<tr><td></td><td>555</td><td>111</td><td>111</td><td>55</td><td>55.5</td><td>Rejected</td></tr>
<tr><td></td><td>57</td><td>11</td><td>11.4</td><td></td><td></td><td>Rejected</td></tr>
<tr><td></td><td>500</td><td>100</td><td>100</td><td>50</td><td>50</td><td></td></tr>
<tr><td>550</td><td>–1</td><td></td><td></td><td></td><td></td><td>550</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
</td><td>6</td></tr>
<tr><td>2(b)</td><td>
One mark per mark point, max two

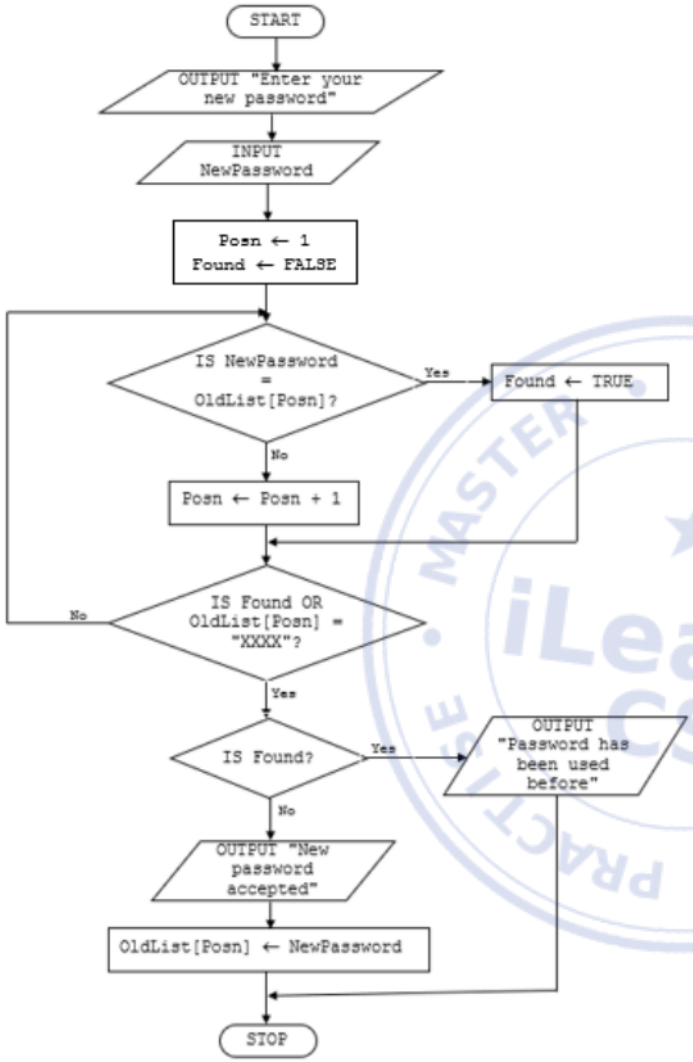
to find if an input is divisible by (both 5 and) 10
… add them together and output the total

</td><td>2</td></tr>
</table>

Question 3 · Oct/Nov 2023 Paper 23, original Q5 (a),(b) [9 marks]

Question	Answer	Marks
3(a)	**One** mark for each error identified and correction given • Line `06 Password` should be `NewPassword` • Line `11 AND` should be `OR` • Line `16 INPUT` should be `OUTPUT`	3

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS

3(b)		**Max six** marks from: **Max four** from: **one** mark for data entry with message **one** mark for initialisation **one** mark for checking list // decision box comparing input with array **one** mark for updating // updating the two variables position and found **one** mark for loop control // second decision box **one** mark for setting new password to position in list **one** mark for outputs // two outputs **Two marks**: **one mark** for correct use of flow chart symbols **one mark** for correct use arrows and labels	**6**

Total: 75 marks

Question 4 · Feb/March 2024 Paper 22, original Q2 (a),(b) [8 marks]

Question	Answer	Marks
4(a)	**One** mark for each correct line from the test data type the description **Test data type** — **Description** abnormal → a value that is accepted boundary → a value that is the highest or lowest value to be accepted and the corresponding lowest or highest value to be rejected extreme → a value that is the highest or lowest value to be rejected normal → a value that is rejected a value that is the highest or lowest value to be accepted	4
4(b)	**One** mark for each point: • Abnormal for example 31 • Boundary 4, 5 // 10, 11 • Extreme 5 // 10 • Normal for example 6	4

Total: 75 marks

Question 5 · May/June 2024 Paper 22, original Q9 (a),(b) [7 marks]

Question	Answer	Marks
5(a)	**One** mark for each of columns **A**, **B** and **T** **Two** marks for columns **List[1]** to **List[5]** all entries correct or **One** mark for columns **List[1]** to **List[5]** with one error	**5**

A	B	List[1]	List[2]	List[3]	List[4]	List[5]	T
		15	17	20	5	9	
FALSE	1	17	15				15
TRUE	2		20	15			15
TRUE	3						
TRUE	4				9	5	5
	5						
FALSE	1	20	17				17
TRUE	2						
	3						
	4						
	5						
FALSE	1						
	2						
	3						
	4						
	5						

5(b)	**One** mark for each point MP1 (bubble) sort data in array MP2 in descending order	**2**

Question 6 · Oct/Nov 2024 Paper 23, original Q5 [6 marks]

Question	Answer	Marks
6	**One** mark for stage, **one** mark for matching description and **one** mark for matching expansion (**max six**) For example: • design (1) construction of a solution (1) using standard methods e.g. flowcharts (1) • coding (1) program is written (1) iterative testing takes place (1) • testing (1) program is tested for errors (1) program is tested that it meets its requirements (1)	6

Question 7 · Feb/March 2025 Paper 22, original Q4 (a),(b) [7 marks]

Question	Answer	Marks
7(a)	To check that a value has been entered.	1
7(b)(i)	Type (check)	1

7(b)(ii)	**One** mark per mark point:
	MP1 Condition controlled loop used
	MP2 Working input and re-input of value into declared variable
	MP3 Use of selection/loop entry/exit condition to test if input is an integer
	MP4 Appropriate use of error message
	MP5 Working complete termination of loop
	
	Example:
	`REPEAT`
	`  INPUT Number`
	`  IF MOD(Number, 1) <> 0`
	`  THEN`
	
	`  OUTPUT "Please try again"`
	`  ENDIF`
	`UNTIL MOD(Number, 1) = 0`
	
	Or
	
	`INPUT Number`
	`WHILE MOD(Number, 1) <> 0 (DO)`
	`  OUTPUT "Please try again"`
	`  INPUT Number`
	`ENDWHILE`
		5

Question 8 · May/June 2025 Paper 22, original Q1 [1 marks]

Question	Answer	Marks
8	D	1

Question 9 · May/June 2025 Paper 23, original Q8 [15 marks]

Question	Answer	Marks
9	Marks are available for: • AO2 (maximum 9 marks) • AO3 (maximum 6 marks) **Data Structures required** with names as given in the scenario: Arrays or lists `Video[]`, `Results[]` **Requirements (techniques)** **R1** initialises arrays, displays menu and allows choice, then proceeds based on valid choice, (nested iteration, input, output, selection, validation). **R2** enters data and stores into array at first available location. The code allows for additional data to be input, as required. (counting, selection, input, storage, iteration if selected). **R3** finds specific data based on user input and outputs results. Appropriate output messages, including if video not found. Program continues until stopped by user. (input, output, linear search, iteration).	**15**

Example 15-mark answer in pseudocode

```
//Array and variable declarations not required in responses
//Initialisation of Video array
FOR Index1 ← 1 TO 10000
    FOR Index2 ← 1 TO 4
        Video[Index1, Index2] ← ""
    Next Index2
NEXT Index1
//Display of menu choices
REPEAT
    OUTPUT "Enter 1 to input data for a new video, 2 to search for a video title, or 3 to stop"
    //Input menu choice with validation of input
    REPEAT
        INPUT Answer
        IF Answer <> 1 AND Answer <> 2 AND Answer <> 3
          THEN
            OUTPUT "You must input 1, 2 or 3, please try again"
        ENDIF
    UNTIL Answer = 1 OR Answer = 2 OR Answer = 3
    //User chooses to input data for a new video title
    IF Answer = 1
      THEN
        Store ← 1
        //Finding the next available space in the array
        REPEAT
            IF Video[Store, 1] <> ""
              THEN
                Store ← Store + 1
            ENDIF
        UNTIL Video[Store, 1] = ""
        //Data entry and storage until user says no more needed
        REPEAT
            OUTPUT "Enter the title"
            INPUT Video[Store, 1]
            OUTPUT "Enter the format (4K for 4K, BD for Blu-ray, DV for DVD, DG for Digital"
            INPUT Video[Store, 2]
            OUTPUT "Enter the year of release"
            INPUT Video[Store, 3]
```

```
            OUTPUT "Enter the storage code"
            INPUT Video[Store, 4]
             Store ← Store + 1
            OUTPUT "Another video (Y or N)?"
            INPUT Another
        UNTIL Another = 'N' OR Another = 'n'
    ENDIF
    //User chooses to search for a video title
    IF Answer = 2
      THEN
        REPEAT
            //Initialisation of Results array
             FOR Index1 ← 1 TO 20
                 FOR Index2 ← 1 TO 4
                     Results[Index1, Index2] ← ""
                NEXT Index2
            NEXT Index1
            //User inputs their search criterion
            OUTPUT "State the title of the video you want"
            INPUT Title
             Search ← 1
             Store ← 1
            //Searching for the title
            REPEAT
                //If found, data copied to Results array
                IF Video[Search, 1] = Title
                  THEN
                     FOR Index1 ← 1 TO 4
                         Results[Store, Index1] ← Video[Search, Index1]
                    NEXT Index1
                    //Search and Store indexes incremented
                     Search ← Search + 1
                     Store ← Store + 1
                  ELSE
                    //Search index incremented
                     Search ← Search + 1
                ENDIF
```

```
            //Search ends when no more data in Video array,
            //or end of Video array reached.
            UNTIL Video[Search, 1] = "" OR Search = 10000
            //Outputting only results found from Results array
            IF Results[1, 1] <> ""
              THEN
                OUTPUT "The results: "
                 FOR Index1 ← 1 TO Store - 1
                     FOR Index2 ← 1 TO 4
                        OUTPUT Results[Index1, Index2]
                    NEXT Index2
                NEXT Index1
              ELSE
              //Feedback to user if item not found
                OUTPUT "Item not found"
            ENDIF
            //Another search offered
            OUTPUT "Another search (Y or N)?"
            INPUT Another
        UNTIL Another = 'N' OR Another = 'n'
    ENDIF
UNTIL Answer = 3
```

Marking Instructions in italics

AO2: Apply knowledge and understanding of the principles and concepts of computer science to a given context, including the analysis and design of computational or programming problems

Question 10 · May/June 2026 Paper 23, original Q5 (a),(b) [7 marks]

Question	Answer	Marks	Guidance
10	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		

10(a)	**One** mark for each correct column. **PassCount** | **FailCount** | **Count** | **Size** | **OUTPUT** 0 | 0 | 0 | | 1 | | 1 | 49.8 | 2 | | 2 | 48.7 | | 1 | 3 | 62.5 | 3 | | 4 | 49.9 | 4 | | 5 | 30.0 | | 2 | 6 | 58.2 | 5 | | 7 | –12.5 | | 3 | 8 | 50.0 | 6 | | 9 | 26.7 | | | | | Number passed: 6 | | | | Number failed: 3	**5**	If no other marks awarded, allow 1 mark for correct initialisation of `PassCount`, `FailCount` and `Count` to 0 Repeated errors such as not initialising all columns, penalise once and FT Ignore case in output, but punctuation must be correct

10(b)	Any **two** from: • Add another decision box after the input box and before the existing decision box. • Check for values <0 inside this decision box • If the condition fails, loop back to the input box for re-input // if the condition passes, go to the next decision box	**2**	This is a flowchart question, so answers must be in the context of a flowchart, or changes to the existing flowchart must be identifiable. Accept answers written as flowchart boxes.

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS